Meeting Record

Q: Apply the Bresenhams line drawing Algo to draw line from $(20, 15)$ to $(30, 30)$

Sol^n: Calculate Slope$(m) = \frac{30-15}{30-20} = \frac{15}{10} = 1.5$ and grater than 1.

if $P_k \geq 0$:

$P_{k+1} = P_k + 2\Delta x - 2\Delta y(x_k + 1 - x_k)$

$\boxed{P_{k+1} = P_k + 2\Delta x - 2\Delta y}$

if $P_k < 0$

$P_{k+1} = P_k + 2\Delta x - 2\Delta y(x_k - x_k)$

$\boxed{P_{k+1} = P_k + 2\Delta x}$

$m > 1$:

y moves in unit interval

i.e. $y_{k+1} = y_k + 1$

$P_k = 2\Delta x - \Delta y$

$P_{k+1} = P_k + 2\Delta x - 2\Delta y(x_{k+1} - x_k)$

if $P_k \geq 0$	if $P_k < 0$
$x_{k+1} = x_k + 1$, $y_{k+1} = y_k + 1$	$x_{k+1} = x_k$, $y_{k+1} = y_k + 1$

Steps	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1	20	15	$P_0 = 5$; $P_{k+1} = P_k + 2\Delta x - 2\Delta y = -5$	21	16
2	21	16	$P_{k+1} = P_k + 2\Delta x = -5 + 20 = 15$	21	17
3	21	17	$P_{k+1} = P_k + 2\Delta x - 2\Delta y = 15 + 20 - 30 = 5$	22	18
4	22	18	$P_{k+1} = P_k + 2\Delta x - 2\Delta y = 5 + 20 - 30 = -5$	23	19
5	23	19	$P_{k+1} = P_k + 2\Delta x = -5 + 20 = 15$	23	20
6	23	20		24	21

$m > 1$

Δy steps same

$30 - 15 = 15$

$(20, 15)$

$P_0 = 2\Delta x - \Delta y$

$\Delta x = 30 - 20 = 10$

$\Delta y = 15$

Q: Apply Bresenhams line drawing Algo to draw a line from m<1

(-3,0) and end point is (4,4).

Solⁿ: Calculate slope(m) = $\frac{4-0}{4-(-3)} = \frac{4}{7} = 0.57 < 1$

Step	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1	-3	0	$P_0 = 8 - 7 = 1$	-2	1
2	-2	1	$P_{k+1} = 1 + 8 - 14 = -5$	-1	1
3	-1	1	$P_{k+1} = -5 + 8 = 3$	0	2
4	0	2	$P_{k+1} = 3 + 8 - 14 = -3$	1	2
5	1	2	$P_{k+1} = -3 + 8 = 5$	2	3
6	2	3	$P_{k+1} = -1$	3	3
7	3	3	$P_{k+1} = 7$	$\boxed{4}$	$\boxed{4}$

if $P_k \geq 0$	if $P_k < 0$
$x_{k+1} = x_k + 1$	$x_{k+1} = x_k + 1$
$y_{k+1} = y_k + 1$	$y_{k+1} = y_k$
$P_0 = 2\Delta y - \Delta x$	$P_0 = 2\Delta y - \Delta x$

$$P_{k+1} = P_k + 2\Delta y - 2\Delta x(y_{k+1} - y_k)$$

$$= P_k + 2\Delta y - 2\Delta x(y_k + 1 - y_k)$$

$$\boxed{P_{k+1} = P_k + 2\Delta y - 2\Delta x}$$

$$\boxed{P_{k+1} = P_k + 2\Delta y}$$